

Finance Dashboard — Calculation Logic Reference

Ye document teeno files (`index.html`, `transactions.html`, `cashflow.html`) ke **saare calculation/formula parts** ko ek jagah list karta hai — file, function name aur (approx.) line number ke saath — taaki future mein koi bhi formula change karna ho to seedha sahi jagah jaake edit kiya ja sake.

TIP: Tip: Har file mein **Ctrl+F** se function ka naam search kar lo, seedha wahin pahunch jaoge.

FILE 1: `index.html` — Main Dashboard

A. Raw Excel data se numbers banana → `buildDataFromWorkbook(wb)`

Ye sabse important function hai — Excel ki **Transactions** sheet padh ke saare KPIs/charts ke liye base data taiyaar karta hai.

Har transaction row ke liye, `type` column ke hisaab se:

```
if (typeNorm === 'income')    { yd.income += amt;    balanceTotal += amt; }
else if (typeNorm === 'expense') { yd.expense += amt;    balanceTotal -= amt; }
else if (typeNorm === 'investment') { yd.investment += amt; balanceTotal -= amt; }
```

Kahan change karna hai agar...

Kya change karna hai	Kahan
Balance calculate karne ka tarika (kaunsa type add/subtract ho)	<code>buildDataFromWorkbook()</code> — upar wala if/else block
Naya transaction type add karna ho (e.g. "Loan")	Same block mein naya <code>else if</code> add karo + <code>ensureYear()</code> mein naya bucket

B. Category totals (Expense add, Income/Investment subtract)

```
const catAmt = (typeNorm === 'expense') ? amt : -amt;
```

- "Salary" aur "Other Income" categories `EXCLUDED_CATS` list mein hain — ye category charts mein kabhi nahi dikhtin. - Change karna ho: `buildDataFromWorkbook()` ke andar `EXCLUDED_CATS` array aur `catAmt` wali line.

C. Monthly breakdown

Har row apne month-index (0-11) ke hisaab se `monthly_income` / `monthly_expense` / `monthly_investment` arrays mein add hoti hai — same function ke andar.

C.1 Column detection (Notes column, Payment ki jagah) → `detectColumns(rows)`

Transactions sheet ke header row mein columns dhoondhne ka logic yahin hai:

```
const wanted = { date:['date'], desc:['description','desc'], type:['type'],
  category:['category'], amount:['amount'], notes:['notes'] };
```

Pehle yahan `payment` (ya "Payment Mode"/"Mode") column detect hota tha aur uska ek alag `payTotals` breakdown bhi banta tha — dono hata diye gaye hain (wo breakdown kahin display nahi hota tha). Ab iski jagah **Notes** column detect hota hai — jaisa Cashflow sheet mein pehle se hota hai — aur har row ke saath `notes` field `allTxns/all` array mein save hoti hai. **Kahan change karna hai:** agar Notes column ka

naam Excel mein alag ho (e.g. "Remarks"), to `detectColumns()` ke `notes: ['notes']` array mein naya alias add karo.

D. Main KPI formulas → `render(year)` function

```
const net = y.income - y.expense - y.investment; // Net Savings
const rate = y.income ? net / y.income : 0; // Savings Rate
const ratio = y.income ? y.expense / y.income : 0; // Expense-to-Income gauge
```

Kahan change karna hai: `render()` function ke shuru mein, jahan `kSavings`, `kRate`, `gaugeNum` set ho rahe hain.

E. YoY (Year-over-Year) delta/growth %

```
const change = prev !== 0 ? (cur - prev) / Math.abs(prev) : 0;
```

Function: `delta(elId, cur, prev)` — `render()` ke andar hi define hai.

Income/Expense/Investment/Savings — chaaron isi function se calculate hote hain.

F. "All Years" (ALL) view ka aggregation

Function: `getViewData(sel)` (line ~369) — jab dropdown mein "All Years" select karte ho, tab ye har saal ka data sum karke ek combined object banata hai. Agar ALL-years view ka formula alag chahiye to yahin change karo.

G. Year-wise summary table + Grand Total row

`render()` function ke end mein (`yearTableBody` wala part):

```
const n = d.income - d.expense - d.investment; // per-year Net
```

aur last mein ek `reduce()` se **Total (Gross)** row banti hai — sabhi saalon ka sum.

H. "Balance in A/C" KPI

Ye Year selector se independent hai — hamesha `DATA.balance` (poore all-time `balanceTotal`) dikhata hai:

```
document.getElementById('kBalance').textContent = fmtINR(DATA.balance || 0);
```

FILE 2: `transactions.html` — All Transactions page

Summary cards → `renderSummary()`

```
// Expense + Investment dono "paisa bahar gaya" (Given) maana jaata hai
const given = sum of (Expense + Investment amounts)
const received = sum of (Income amounts)
const net = received - given
```

Kahan change karna hai: `renderSummary()` function — agar Investment ko "given" na maankar alag rakhna ho, to yahan ka condition badlo.

Table ke neeche wala running "Net" total → `renderPage()`

```
const netTotal = sum of (Expense/Investment = negative, Income = positive)
```

Same logic, dusri jagah use hota hai (filtered rows ke liye).

Filters (year/month/type/category/search) → `applyFilters()`

Ye sirf filtering hai, calculation nahi — lekin agar naya filter add karna ho to yahin add hoga.

Notes column (Payment ki jagah) → `renderPage()`

Table ka 5th column ab `t.notes` display karta hai (pehle `t.payment` tha, jo hata diya gaya hai). Ye sirf display hai — koi calculation nahi. Actual Notes text **Excel workbook se hi aata hai** (Transactions sheet ka "Notes" column) — `index.html` ke `buildDataFromWorkbook()` mein parse hota hai, phir `data.js/data.json` mein save hota hai jise ye page read karta hai.

FILE 3: `cashflow.html` — Lender Ledger

KPI cards → `renderKPIs()`

```
Given      = sum of amount where type === 'Given'
Received   = sum of amount where type === 'Received'
Net Balance = Given - Received
```

Ye alag hi Excel sheet ("Cashflow") se aata hai — Transactions sheet se koi lena-dena nahi. Iska raw parsing `index.html` ke andar `parseCashflowSheet(wb)` function mein hota hai (data yahin se generate hoke `data.js` / `data.json` mein save hota hai).

Table ke neeche "Net" total → `renderPage()`

Same formula, filtered rows ke liye.

Quick Cheat-Sheet — "Mujhe X change karna hai, kahan jaaun?"

Change karna hai	File	Function
Balance kaise calculate ho (kaunsa type +/-)	<code>index.html</code>	<code>buildDataFromWorkbook()</code>
Net Savings ka formula	<code>index.html</code>	<code>render()</code>
Savings Rate %	<code>index.html</code>	<code>render()</code>
Category chart mein kaun-si categories exclude hon	<code>index.html</code>	<code>buildDataFromWorkbook()</code> → <code>EXCLUDED_CATS</code>
Expense/Investment "paisa bahar" maana jaaye ya nahi	<code>transactions.html</code>	<code>renderSummary()</code> , <code>renderPage()</code>
Lender-wise Given/Received/Net	<code>cashflow.html</code>	<code>renderKPIs()</code>
Cashflow sheet se data kaise parse ho	<code>index.html</code>	<code>parseCashflowSheet()</code>
YoY growth % ka formula	<code>index.html</code>	<code>delta()</code> (inside <code>render()</code>)
"All Years" view ka total kaise bane	<code>index.html</code>	<code>getViewData('ALL')</code>
Transactions sheet mein Notes column ka naam/alias	<code>index.html</code>	<code>detectColumns()</code> → <code>notes: ['notes']</code>

Change karna hai	File	Function
Transactions table mein Notes column display	transactions.html	<code>renderPage()</code>

NOTE: Ek important baat

Actual numbers **Excel workbook se** `buildDataFromWorkbook()` (**index.html**) mein hi calculate hote hain, aur phir `data.json` / `data.js` mein save ho jaate hain. `transactions.html` aur `cashflow.html` sirf us already-calculated data ko **read/filter/display** karte hain — wahan naye totals sirf filtered subset ke liye bante hain (jaise summary cards), raw Excel data wahan nahi padha jaata.

Isliye: agar **core calculation logic** (balance, category rule, income/expense type) change karna hai → hamesha **index.html** mein jaake `buildDataFromWorkbook()` edit karo, aur phir Excel wapas load/push karo taaki `data.js/data.json` refresh ho jaaye.

Protected-pages password (transactions.html / cashflow.html) — online se change karna

Ye calculation nahi hai, lekin GitHub Sync jaisi hi push mechanism use karta hai isliye yahan note kiya gaya hai.

- Dashboard (**index.html**) ke ■■ **GitHub Sync** panel mein ek naya section hai: "**Change Protected-Pages Password**". - Naya password type karo → **Update Password** click karo. - Function: `updatePasswordOnGithub()` — same `pushOneFileToGithub()` reuse karta hai jo `data.json/data.js` push karne ke liye already istemal hota hai, sirf path `auth-config.js` hai. Content `authConfigContent(newPassword)` se banta hai (same file jaisa comments-wala format, sirf **PASSWORD** value badalti hai). - **Requirement:** GitHub Sync (repo + token, Contents: Read & write) pehle se set-up hona chahiye — same setup jo data push ke liye use hota hai. - Push hone ke baad naya password 1-2 minute mein live ho jaata hai (GitHub Pages ke build/CDN cache ke hisaab se); purana password tab kaam nahi karega. - **Kahan change karna hai agar format badalna ho:** **index.html** → `authConfigContent()` function.